SYLLABUS : ROTATIONAL BODY DYNAMICS

1. On account of the earth rotating about its axis :-

(A) the linear velocity of objects at equator is greater than at other places

(B) the angular velocity of objects at equator is more than that of objects at poles

(C) the linear velocity of objects at all places at the earth is equal, but angular velocity is different

(D) at all places the angular velocity and linear velocity are uniform

2. A fly wheel originally at rest is to reach an angular velocity of 36 radian/s in 6 second. The total angle it turns through in the 6 second is

(A) 54 radian (B) 108 radian (C) 6 radian (D) 216 radian

3. The rotating rod starts from rest and acquires a rotational speed n = 600 revolution/minute in 2 seconds with constant angular acceleration. The angular acceleration of the rod is

(A) $10\ \pi$ rad/s^2 (B) $5\ \pi$ rad/s^2 (C) $15\ \pi$ rad/s^2 (D) None of these

4. A fan is running at 3000 rpm. It is switched off. It comes to rest by uniformly decreasing its angular speed in 10 seconds. The total number of revolutions in this period.

(A) 150 (B) 250 (C) 350 (D) 300

5. A sphere is rotating uniformly about an axis passing through its centre then:

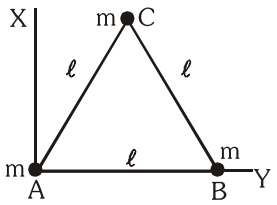
(A) The particles on the surface of the sphere do not have any angular acceleration.

(B) The particles on the axis do not have any linear acceleration

(C) Different particles on the surface have same angular speeds.

(D) All the particles on the surface have same linear speed

6. Three particles, each of mass m are situated at the vertices of an equilateral triangle ABC of side ℓ cm (as shown in the figure). The moment of inertia of the system about a line AX perpendicular to AB and in the plane of ABC, in gram cm^2 units will be :–



(A) $2\ m\ell^2$ (B) $\frac{5}{4}m\ell^2$ (C) $\frac{3}{2}m\ell^2$ (D) $\frac{3}{4}m\ell^2$

7. A circular disc is to be made by using iron and aluminium so that it acquired maximum moment of inertia about geometrical axis. It is possible with :–

(A) aluminium at interior and iron surrounded to it.

(B) iron at interior and aluminium surrounded to it.

(C) using iron and aluminium layers in alternate order.

(D) sheet of iron is used at both external surface and aluminium sheet as internal layer.

8. The moment of inertia of a thin uniform rod of mass M and length ℓ about an axis perpendicular to the rod, through its centre is I. The moment of inertia of the rod about an axis perpendicular to the rod through its end point is :–

(A) I/4 (B) I/2 (C) 2I (D) 4I

9. Two rings of the same radius and mass are placed such that their centres are at a common point and their planes are perpendicular to each other. The moment of inertia of the system about an axis passing through the centre and perpendicular to the plane of one of the rings is (mass of the ring = m, radius = r) :–

(A) $\frac{1}{2}mr^2$ (B) mr^2 (C) $\frac{3}{2}mr^2$ (D) $2mr^2$

10. A uniform circular disc A of radius r is made from A copper plate of thickness t and another uniform circular disc B of radius 2r is made from a copper plate of thickness t/2. The relation between the moments of inertia I_A and I_B is

(A) $I_A > I_B$ (B) $I_A = I_B$

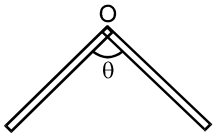
(C) $I_A < I_B$ (D) depends on the values of t and r.

11. Let I_A and I_B be the moments of inertia of two solid cylinders of identical geometrical shape and size about their axes, the first made of aluminium and the second of iron.

(A) $I_A < I_B$ (B) $I_A = I_B$ (C) $I_A > I_B$

(D) relation between I_A and I_B depends on the actual shapes of the bodies.

12. A unifrom thin rod of length L and mass M is bent at the middle point O as shown in figure. Consider an axis passing through its middle point O and perpendicular to the plane of the bent rod. Then moment of inertia about this axis is :



(A) $\frac{2}{3}mL^2$ (B) $\frac{1}{3}mL^2$ (C) $\frac{1}{12}mL^2$ (D) dependent on θ

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13. If a rigid body is subjected to two forces $\vec{F}_1 = 2\hat{i}+3\hat{j}+4\hat{k}$ acting at (3,3,4) and $\vec{F}_2 = -2\hat{i}-3\hat{j}-4\hat{k}$ acting at (1, 0, 0) then which of the following is (are) true?

(A) The body is in equilibrium.

(B) The body is under the influence of a torque only.

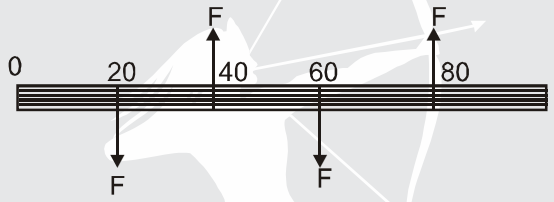
(C) The body is under the influence of a single force.

(D) the body is under the influence of a force together with a torque .

14. A force $\vec{F} = 2\hat{i}+3\hat{j}-\hat{k}$ acts at a point (2, -3, 1). Then magnitude of torque of this force about point (0, 0, 2) will be:

(A) 6 (B) $3\sqrt{5}$ (C) $6\sqrt{5}$ (D) none of these

15. Four equal and parallel forces are acting on a rod (as shown in figure) in horizontal plane at distances of 20 cm, 40 cm, 60 cm and 80 cm respectively from one end of the rod. Under the influence of these forces the rod :



(A) is at rest (B) experiences a torque

(C) experiences a linear motion (D) experiences a torque and also a linear motion

16. A rod of weight w is supported by two parallel knife edges A & B and is in equilibrium in a horizontal position. The knives are at a distance d from each other. The centre of mass of the rod is at a distance *x* from A. The normal reactions at A and B will be :

(A) $N_A = 2w\ (1 - x/d)$, $N_B = wx/d$] (B) $N_A = w\ (1 - x/d)$, $N_B = wx/d$]

(C) $N_A = 2w\ (1 - x/d)$, $N_B = 2wx/d$] (D) $N_A = w\ (2 - x/d)$, $N_B = wx/d$]

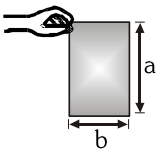
17. A wheel is rotating about an axis through its centre at 720 rpm. It is acted on by a constant torque opposing its motion for 8 second to bring it to rest finally. The value of torque in Nm is :– (given $I = \frac{24}{\pi} kg-m^2$)

(A) 48 (B) 72 (C) 96 (D) 120

18. A rod of mass M and length L is placed in a horizontal plane with one end hinged about the vertical axis. A horizontal force of $F=\frac{Mg}{2}$ is applied at a distance $\frac{5L}{6}$ from the hinged end. The angular acceleration of the rod will be :-

(A) $\frac{4g}{5L}$ (B) $\frac{5g}{4L}$ (C) $\frac{3g}{4L}$ (D) $\frac{4g}{3L}$

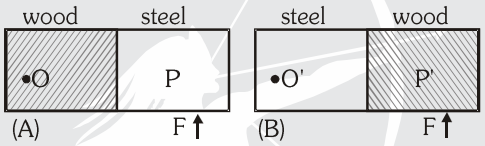
19. A person supports a book between finger and thumb as shown (the point of grip is assumed to be at the corner of the book). If the book has a weight of W then the person is producing a torque on the book of



(A) W $\frac{a}{2}$ anticlockwise (B) W $\frac{b}{2}$ anticlockwise

(C) Wa anticlockwise (D) Wa clockwise

20. In the figure (A) half of the meter scale is made of wood while the other half of steel. The wooden part is pivoted at O. A force F is applied at the end of steel part. In figure (B) the steel part is pivoted at O' and the same force is applied at the wooden end:–



(A) More angular acceleration will be produced in (A)

(B) More angular acceleration will be produced in (B)

(C) Same angular acceleration will be produced in both conditions

(D) Information is incomplete

21. A uniform circular disc A of radius r is made from a metal plate of thickness t and another uniform circular disc B of radius 4r is made from the same metal plate of thickness t/4. If equal torques act on the discs A and B, initially both being at rest. At a later instant, the angular speeds of a point on the rim of A and another point on the rim of B are ω_A and ω_B respectively. We have

(A) $\omega_A > \omega_B$ (B) $\omega_A = \omega_B$ (C) $\omega_A < \omega_B$

(D) the relation depends on the actual magnitude of the torques.

22. One end of a uniform rod having mass m and length ℓ is hinged. The rod is placed on a smooth horizontal surface and rotates on it about the hinged end at a uniform angular velocity ω. The force exerted by the hinge on the rod has a horizontal component

(A) $m\omega^2\ell$ (B) zero (C) mg (D) $\frac{1}{2}m\omega^2\ell$

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23. Two men support a uniform horizontal rod at its two ends. If one of them suddenly lets go, the force exerted by the rod on the other man just after this moment will:

(A) remain unaffected

(B) increase

(C) decrease

(D) become unequal to the force exerted by him on the rod.

24. Two rotating bodies have same angular momentum but their moments of inertia are I_1 and I_2 respectively ($I_1 > I_2$). Which body will have higher kinetic energy of rotation:–

(A) First (B) Second

(C) Both will have same kinetic energy (D) Not possible to predict

25. A thin rod of length L and mass m is suspended from one end and rotated with n rotations per second. The rotational kinetic energy of the rod will be:–

(A) $2mL^2\pi^2n^2$ (B) $\frac{1}{2}mL^2\pi^2n^2$ (C) $\frac{2}{3}mL^2\pi^2n^2$ (D) $\frac{1}{6}mL^2\pi^2n^2$

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ANSWER KEY

1.	(A)	**2.**	(B)	**3.**	(A)	**4.**	(B)	**5.**	(AC)
6.	(B)	**7.**	(A)	**8.**	(D)	**9.**	(C)	**10.**	(C)
11.	(A)	**12.**	(C)	**13.**	(A)	**14.**	(C)	**15.**	(B)
16.	(B)	**17.**	(B)	**18.**	(B)	**19.**	(B)	**20.**	(B)
21.	(A)	**22.**	(D)	**23.**	(C)	**24.**	(B)	**25.**	(C)